

Kim Yungju

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Aug 2024 - Present

Bachelor of Computing (Computer Science), Honours

- Transferred from BBA & Economics to Computer Science to pursue technical specialization
- Expected Graduation: DEC 2027

EXPERIENCE

LUMCLOON ENERGY | Data Center Systems Intern

Dec 2025 - Jan 2026

- Authored a 46-page internal assessment on Ireland's new Large Energy User (LEU) framework, analyzing the transition of the data center market into a regulatory-driven maturity stage.
- Modeled grid capacity constraints and de-rating mechanisms for hyperscale deployment, evaluating the integration of SOFC (160MW), BESS (250MW), and CPPA renewables.
- Identified a 73MW capacity deficit by applying CRM de-rating curves and formulated a Credible Plan for 80% renewable matching, securing the technical foundation for a compliant grid connection application.

CO-CURRICULAR ACTIVITIES

Developer Group @ NUS Computing | Head of Technology

Aug 2025 - Present

- Contributed to a student-led tech community (2,300+ followers) connecting communities with technology.
- Designed and developed the organization's website to showcase events, initiatives, and partner collaborations, improving visibility and outreach.
- Organized and managed the technical operations of Hack4Good 2026, a flagship hackathon designing impactful solutions for non-profit organizations.

PROJECTS

Colonial Archives Graph-RAG — Singapore History + AI Hackathon, Top 3

- Built a full-stack AI research tool that processes colonial-era archive PDFs (OCR → chunking → embedding → entity extraction → knowledge graph construction), enabling source-grounded Q&A with every answer traced to a specific document page.
- Engineered a hybrid RAG retrieval system combining Vertex AI Vector Search and Neo4j graph traversal, scoring results via weighted relevance and generating cited answers with Gemini 2.0 Flash.
- Built with FastAPI, React 19, Google Cloud (Cloud Storage, Document AI, Vertex AI, Gemini 2.0 Flash), Neo4j AuraDB, and Cytoscape.js.

PriceWise — AI Shopping Agent

- Developed an autonomous product research agent that compares prices and reviews across retailers, delivering structured receipts via a conversational interface with selective human-in-the-loop approval.
- Engineered a LangGraph agent with 10 orchestrated tools, dual-mode SSE streaming, persistent checkpointing via AsyncPostgresSaver, and conversation summarization with safe message splitting.
- Built with LangGraph, FastAPI, OpenAI, Tavily, Pydantic, PostgreSQL, Next.js, and Docker, deployed on Railway (backend) and Vercel (frontend).

InterviewPilot — AI Mock Interview Platform

- Built a full-stack mock interview platform that generates role-tailored questions, conducts speech-based interviews with webcam recording, and delivers AI-scored feedback.
- Implemented multi-turn conversational flow with AI-generated follow-up questions, fire-and-forget video upload pipeline with late URL patching, and cross-browser MediaRecorder MIME negotiation.
- Built with Next.js, React, TypeScript, OpenAI, Drizzle ORM, Supabase, and Clerk, with bilingual support (English/Korean) including dual TTS pipeline and CJK PDF export.

Castory — AI Podcast Platform

- Developed a full-stack SaaS platform enabling users to generate AI-narrated podcasts from trending news via a 5-step guided workflow using OpenAI.
- Implemented chunked TTS pipeline, persistent audio player, full-text search, draft auto-save, and dynamic user profiles with listener analytics.
- Built with Next.js, React, TypeScript, Convex (serverless backend + real-time DB), and Clerk (webhook-synced auth), deployed on Vercel.

SKILLS AND OTHER INFORMATION

- Programming Languages: JavaScript, Java, Python, and SQL
- Frameworks & Tools: Next.js, React, LangGraph, FastAPI, Docker, PostgreSQL, Supabase
- Fluent in written and spoken English and Korean